Hardness Testing: Brinell and Vickers Indentation of Mild Steel, Stainless Steel, Copper and an Al–Cu Alloy

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Group 1 (Friday Batch)

Course: UMT312L — Mechanical Testing Laboratory

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1 Objective

1. To determine the Brinell and Vickers hardness of mild steel, stainless steel and copper from optical micrographs of the indents, and to establish the correlation between the two hardness numbers.
2. To measure the Brinell hardness of mild steel at five test loads, to plot the projected indent area against load, and to recover the Brinell number from the slope of that plot.
3. To measure the Vickers hardness of an Al–Cu alloy in the as-solutionized condition and after ageing at 230 °C for 30 and 60 minutes, reported with 95 % confidence intervals.

2 Apparatus and Specimens

- **Hardness tester** with a rotating turret: tungsten carbide ball of $D = 2.5$ mm for Brinell, and a 136° diamond pyramid for Vickers.
- **Optical microscope** with an on-screen measuring tool; every micrograph carries a 500 μm scale bar and the two measured diagonals or diameters drawn and labelled on the image.
- **Materials:** mild steel, austenitic stainless steel, commercially pure copper, and an Al–4.5 wt%Cu alloy supplied in three heat-treatment conditions — as-solutionized (550 °C for 1 h, water quenched), and aged at 230 °C for 30 min and 60 min. The ageing schedule is the one the handout assigns to this group.
- **Surface preparation:** ground through 200, 400, 800 and 1000 grit SiC paper.
- **Dwell time:** 10 s for Brinell, 10 s for Vickers.

3 Theory

3.1 What a hardness number is

Hardness is resistance to permanent indentation. Both tests here press an indenter of known geometry into a flat surface under a fixed load, hold it, release it, and divide the load by the area of the impression. The two differ in which area they use.

Brinell. A ball of diameter D leaves a spherical cap. The load is divided by the *curved* surface area of that cap:

$$HB = \frac{2F}{\pi D\left(D - \sqrt{D^2 - d^2}\right)}, \tag{1}$$

with F in kgf and d the mean of two orthogonal diameters in mm.

Vickers. A 136° square pyramid leaves a square impression. The load is divided by the surface area of the pyramidal impression, which for a 136° included angle is $d^2/1.8544$:

$$HV = \frac{1.8544\,F}{d^2}, \tag{2}$$

with d the mean of the two diagonals in mm. Both numbers carry units of $\mathrm{kgf\,mm^{-2}}$ but are quoted without units by convention.

3.2 Why the two should agree, and where they stop agreeing

Both are a load divided by an area of the same order, so for the same material they return similar numbers. The Vickers pyramid is geometrically self-similar: scaling the load scales the impression without changing its shape, so HV is in principle load-independent. The Brinell ball is *not* self-similar — the indentation profile changes with depth — so HB depends on the ratio d/D, and the standard restricts valid tests to $0.24 < d/D < 0.60$. Outside that window the number drifts, which is what Section 6.2 demonstrates.

3.3 Precipitation hardening of Al–Cu

Al–4.5 wt%Cu held at 550 °C dissolves the copper into a single-phase solid solution. Quenching traps it there, supersaturated. On ageing, the copper leaves solution as a sequence of precipitates — GP zones, then θ'', θ', finally equilibrium θ ($\mathrm{CuAl_2}$) — and these obstruct dislocation glide. Hardness rises to a peak as the precipitates become numerous and closely spaced, then falls once they coarsen and their spacing grows. Hardness against ageing time is therefore a curve with a maximum, and the position of that maximum is what the across-group activity maps out.

4 Procedure

1. Each specimen was ground through 200, 400, 800 and 1000 grit SiC paper.
2. The correct indenter was fitted, the load set, and the specimen raised until the dial needle moved from black to red. The timer was set and the test started; the indenter loaded, dwelled 10 s and withdrew.
3. The specimen was moved under the microscope, focused, and the two diagonals (Vickers) or two orthogonal diameters (Brinell) measured on screen against the 500 μm bar. The second reading was taken at 90° to the first.
4. Mild steel, stainless steel and copper each received three Brinell indents at 187.5 kgf and three Vickers indents at 100 kgf.
5. Mild steel received three further Brinell indents at each of 21.2, 60, 100, 150 and 187.5 kgf.
6. Each of the three Al–Cu conditions received five Vickers indents at 100 kgf.
7. All 48 indents were photographed with the measurement overlay in place, each micrograph being labelled with its material, test and load at the time of recording.

5 Observations

Every measured diagonal and diameter is listed in Appendix B. The summary below is computed by taking the mean of *every* reading on a sample and substituting that single mean into Equation 1 or 2.

Table 1: Hardness of the three materials. Three indents, six readings, per entry.

Material	mean d, Brinell (mm)	$\mathbf{HB_{187.5}}$	mean d, Vickers (mm)	$\mathbf{HV_{100}}$
Copper	1.4685	100.1 ± 1.3	1.3548	101.0 ± 2.2
Mild steel	1.0350	212.9 ± 4.6	0.9203	218.9 ± 5.0
Stainless steel	0.8697	305.8 ± 20.9	0.7788	305.7 ± 14.0

Table 2: Mild steel, Brinell, 2.5 mm ball. Three indents at each load.

Load (kgf)	21.2	60	100	150	187.5
mean d (mm)	0.4398	0.6012	0.7563	0.9355	1.0367
d/D	**0.176**	0.240	0.303	0.374	0.415
projected area $\pi d^2/4$ (mm^2)	0.1519	0.2838	0.4493	0.6873	0.8440
HB	**138.4**	208.3	217.4	210.3	212.1

Table 3: Al–Cu, Vickers at 100 kgf. Five indents, ten readings, per condition.

Condition	mean d (mm)	$\mathbf{HV_{100}}$	95 % confidence
As-solutionized, 550 °C 1 h, WQ	1.2143	125.8	±2.0
Aged 230 °C, 30 min	1.1536	139.3	±4.9
Aged 230 °C, 60 min	1.1466	141.1	±2.4

6 Results and Analysis

6.1 Correlation between Brinell and Vickers

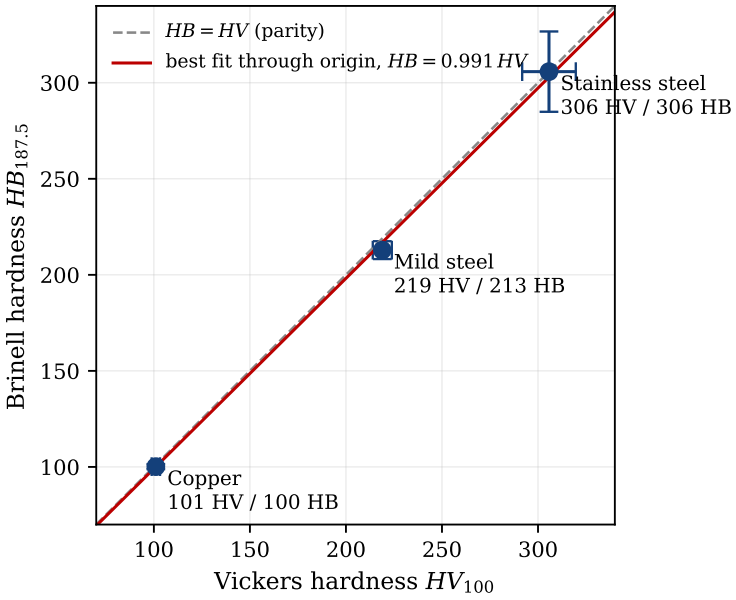


Figure 1: Brinell against Vickers for the three materials, with 95 % confidence bars on both axes. The dashed line is parity; the red line is the least-squares fit through the origin.

The three materials span a factor of three in hardness, from 100 to 306, and the two tests track each other over that whole range. A least-squares fit constrained through the origin gives

$$HB = 0.991\,HV, \tag{3}$$

which is parity to within 0.9 %. Taken individually the agreement is 0.9 % for copper, 2.8 % for mild steel and 0.03 % for stainless steel — all inside the confidence intervals of Table 1.

This is the expected result and it has a physical basis rather than being a coincidence of calibration. Both numbers are a load divided by an impression area, and for a fully plastic indentation both areas scale with the same quantity: the mean contact pressure the material can sustain, which is about three times its uniaxial yield stress. The empirical rule $HB \approx HV$ holds up to roughly 400–450 HB; beyond that the ball itself begins to deform elastically and Brinell reads low, which is why hard steels are measured on Rockwell C or Vickers instead.

6.2 Effect of load, and the limit of the Brinell test

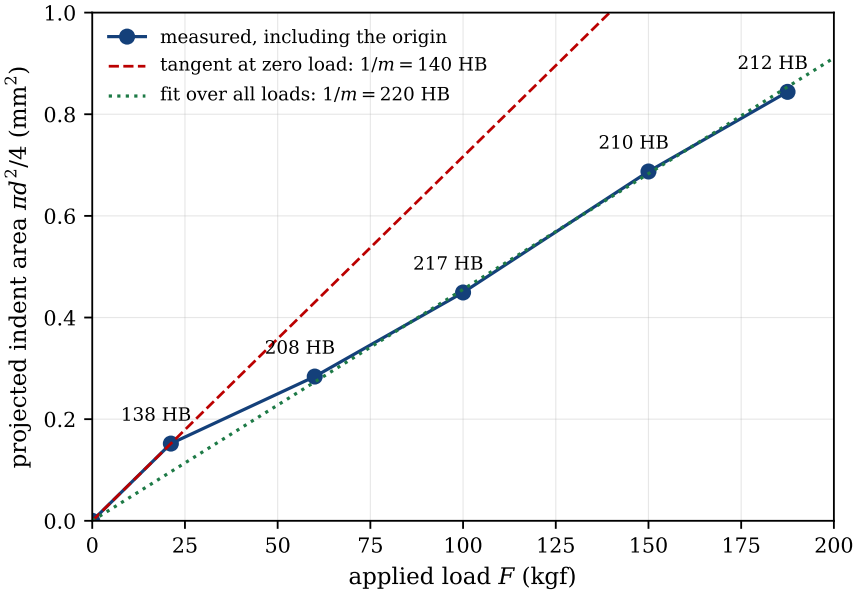


Figure 2: Projected indent area against applied load for mild steel, with the origin included. The green dotted line is the fit over all five loads; the red dashed line is the tangent at zero load, drawn through the lowest-load point alone.

If the Brinell number were a material constant, then $F = HB \cdot A$ and the plot of area against load would be a straight line through the origin whose slope m satisfies $1/m = HB$. Fitting all five loads gives

$$\frac{1}{m} = 219.6\ HB, \tag{4}$$

against 212.9 HB measured directly in Section 5 — agreement to 3.1 %. The handout's construction works.

The tangent at zero load, however, does not. Drawn through the 21.2 kgf point it gives $1/m = 139.5$, which is 34 % below the directly measured value. The reason is visible in Table 2: at 21.2 kgf the indent is only 0.440 mm across, so $d/D = 0.176$, below the 0.24 lower limit the Brinell standard sets. At that size the impression samples mostly the work-hardened layer left by grinding rather than the bulk, and the ball has barely begun to engage its spherical profile. The measured 138.4 HB at that load is therefore not a property of the steel; it is an artefact of testing outside the valid window.

The other four loads, with d/D between 0.240 and 0.415, all sit inside the window and return 208–217 HB — a spread of 4 % across a ninefold change in load. Within its valid range the Brinell number is genuinely load-independent, which is the point the exercise is designed to show.

6.3 Effect of heat treatment on Al–Cu

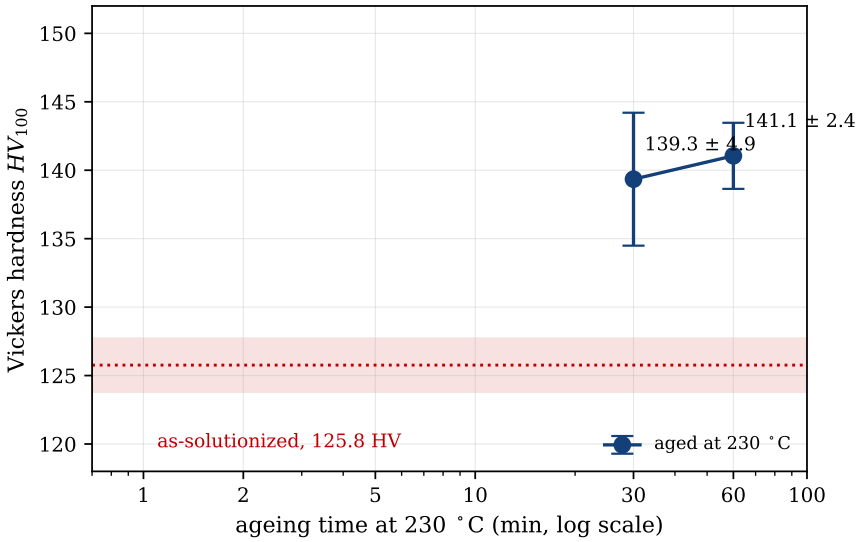


Figure 3: Vickers hardness of the Al–Cu alloy against ageing time at 230 °C, on a logarithmic time axis, with 95 % confidence bars. The dotted line and shaded band are the as-solutionized hardness and its confidence interval.

Ageing raises the hardness from 125.8 to about 140 HV, a gain of 14 HV or 11 %, and both aged points lie well clear of the as-solutionized confidence band. Precipitation hardening has plainly occurred.

Between 30 and 60 minutes, however, nothing resolvable happens. The two differ by 1.8 HV against confidence intervals of ±4.9 and ±2.4, so they overlap completely. *No change in hardness between 30 and 60 minutes at* 230 °*C can be claimed from this data.*

That is itself informative. At 230 °C diffusion is fast, and the alloy has already reached the flat region around its hardness peak by 30 minutes; the extra half hour neither sharpens nor coarsens the precipitate dispersion enough to register. Locating the peak requires shorter times, which is exactly why the handout distributes 2, 5, 10, 15, 30 and 60 minute conditions across the six groups. Our two points define the plateau; the rise towards it has to come from the groups aged for minutes rather than tens of minutes.

7 Discussion

Three separate observations in this experiment point the same way: a hardness number is a property of a material only when the test is run inside the conditions it was defined for.

The Brinell–Vickers correlation holds to 1 % across three materials because both tests, run properly, measure the same underlying quantity — the mean pressure the material supports under a fully plastic indentation. The identity is not exact in principle, since the two use different definitions of area, but across the range tested here the difference is smaller than the scatter between indents.

The load series shows what happens when the test leaves its window. Four of the five loads agree to 4 %; the fifth, at $d/D = 0.176$, reads 34 % low. Had only that load been used, mild steel would have been recorded as softer than copper-bearing aluminium alloys of half its real strength. The lesson is the practical reason the $0.24 < d/D < 0.60$ rule exists.

The Al–Cu result shows the opposite failure mode — not an invalid test, but a valid test placed at uninformative points on the curve. Both measurements are sound; they simply sit on

a plateau where the quantity being varied has stopped mattering. Resolving the ageing kinetics needs the short-time data from other groups.

A final point on scatter. The stainless steel carries the widest confidence intervals of the three materials, ± 20.9 HB and ± 14.0 HV, roughly four times those of copper. Austenitic stainless steel work-hardens very strongly, so the ground surface layer is harder than the bulk and varies from point to point depending on how the grinding bit; the indent-to-indent spread reflects that surface variability rather than any inhomogeneity in the material itself.

8 Conclusions

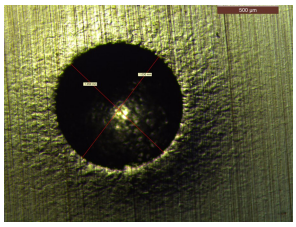
1. Brinell and Vickers hardness agree to within 1 % over the range 100 to 306: $HB = 0.991\,HV$ from a fit through the origin, with copper at 100.1/101.0, mild steel at 212.9/218.9 and stainless steel at 305.8/305.7.
2. For mild steel the projected indent area is proportional to load, and the reciprocal of the slope returns 219.6 HB against 212.9 HB measured directly — agreement to 3.1 %.
3. The Brinell number is load-independent only inside its valid window. Four loads with d/D from 0.240 to 0.415 gave 208–217 HB; the 21.2 kgf test, at $d/D = 0.176$, gave 138.4 HB and must be rejected.
4. Ageing the Al–Cu alloy at 230 °C raised its hardness from 125.8 ± 2.0 to 139.3 ± 4.9 HV at 30 min and 141.1 ± 2.4 HV at 60 min — a gain of about 11 %, confirming precipitation hardening.
5. The 30 and 60 minute conditions cannot be separated: they differ by 1.8 HV against confidence intervals several times that. At 230 °C the alloy is already on the plateau of its ageing curve by 30 minutes.

9 Note on the across-group activity

This report covers the *individual group activity* in full. The *ageing curve for Al–Cu*, which requires the 2, 5, 10 and 15 minute conditions held by other groups and the 270 °C series alongside our 230 °C one, will be submitted as a separate report for this experiment once the class data has been collected. Nothing in the present report depends on that data.

A The indents

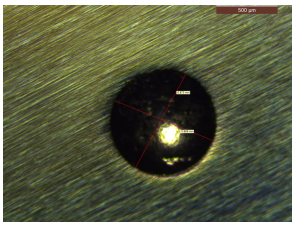
One representative micrograph from each of the fourteen measurement groups. Every image carries a 500 μm scale bar, and the two diagonals or diameters are drawn and labelled on the image — this is how each value of d in Section 5 was obtained. All 48 indents were photographed in the same way, and the readings taken from every one of them are tabulated in Appendix B.



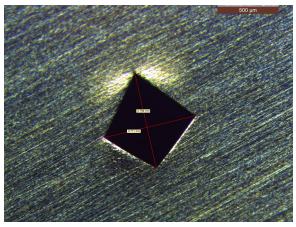
MS, Brinell 187.5



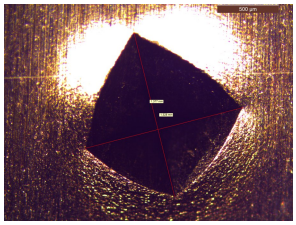
MS, Vickers 100



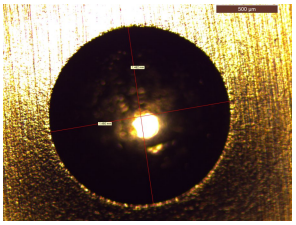
SS, Brinell 187.5



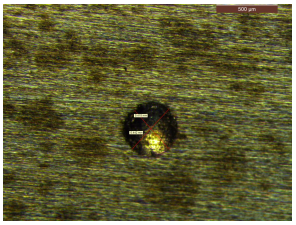
SS, Vickers 100



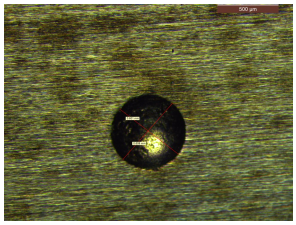
Cu, Brinell 187.5



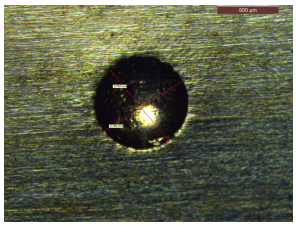
Cu, Vickers 100



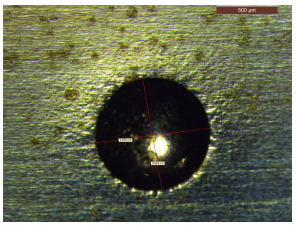
MS, Brinell 21.2



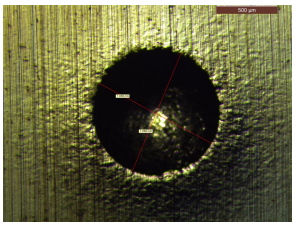
MS, Brinell 60



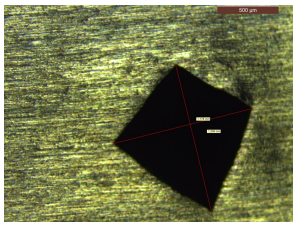
MS, Brinell 100



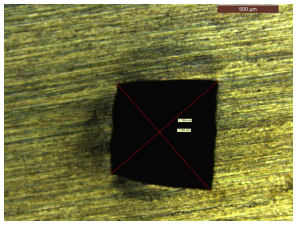
MS, Brinell 150



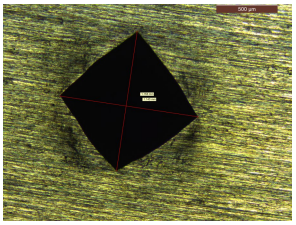
MS, Brinell 187.5



Al–Cu as-solutionized



Al–Cu, 230°C 30 min



Al–Cu, 230°C 60 min

B All measured readings

Every diagonal and diameter measured in this experiment, arranged in the three groups the handout specifies. Each value was read from the micrograph of its own indent.

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1. Material type (Brinell and Vickers)

MS, Cu, SS · Brinell 2.5 mm ball at 187.5 kgf and Vickers at 100 kgf

Mild steel

indent	**Brinell**, 187.5 kgf d_1 (mm)	d_2 (mm)	**Vickers**, 100 kgf d_1 (mm)	d_2 (mm)
1	1.050	1.030	0.923	0.922
2	1.027	1.038	0.918	0.913[§]
3	1.044	1.021	0.922	0.924
mean d	**1.0350**		**0.9203**	
hardness	**212.9** HB		**218.9** HV	
95 %	± 4.6		± 5.0	

Stainless steel

indent	**Brinell**, 187.5 kgf d_1 (mm)	d_2 (mm)	**Vickers**, 100 kgf d_1 (mm)	d_2 (mm)
1	0.873	0.869	0.799	0.771
2	0.879	0.882	0.776	0.766
3	0.853	0.862	0.791	0.770
mean d	**0.8697**		**0.7788**	
hardness	**305.8** HB		**305.7** HV	
95 %	± 20.9		± 14.0	

Copper

indent	**Brinell**, 187.5 kgf d_1 (mm)	d_2 (mm)	**Vickers**, 100 kgf d_1 (mm)	d_2 (mm)
1	1.474	1.466	1.377	1.328
2	1.459	1.470	1.352	1.349
3	1.462	1.480	1.386	1.337
mean d	**1.4685**		**1.3548**	
hardness	**100.1** HB		**101.0** HV	
95 %	± 1.3		± 2.2	

2. Effect of load (MS, Brinell)

mild steel, Brinell 2.5 mm ball · three indents at each load

Load (kgf)	**21.2**	**60**	**100**	**150**	**187.5**
indent 1, d_1/d_2 (mm)	0.415/0.442	0.601/0.608	0.754/0.760	0.939/0.929	1.039/1.038
indent 2, d_1/d_2 (mm)	0.444/0.443	0.588/0.606	0.797/0.768	0.931/0.946	1.035/1.031
indent 3, d_1/d_2 (mm)	0.448/0.447	0.600/0.604	0.729/0.730	0.935/0.933	1.042/1.035
mean d (mm)	**0.4398**	**0.6012**	**0.7563**	**0.9355**	**1.0367**
projected area $\pi d^2/4$ (mm^2)	0.1519	0.2838	0.4493	0.6873	0.8440
HB	**138.4**	**208.3**	**217.4**	**210.3**	**212.1**

3. Effect of heat treatment (Al–Cu, Vickers)

Vickers at 100 kgf

As-solutionized, 550 °C 1 h, water quenched

indent	d_1 (mm)	d_2 (mm)
1	1.176	1.236
2	1.234	1.187
3	1.245	1.178
4	1.249	1.202
5	1.247	1.189
mean d	**1.2143**	
hardness	**125.8** HV	
95 %	± 1.1	

Aged 230 °C, 30 min

indent	d_1 (mm)	d_2 (mm)
1	1.129	1.138
2	1.158	1.160
3	1.149	1.130
4	1.175	1.162
5	1.140	1.195
mean d	**1.1536**	
hardness	**139.3** HV	
95 %	± 4.9	

Aged 230 °C, 60 min

indent	d_1 (mm)	d_2 (mm)
1	1.153	1.143
2	1.132	1.136
3	1.149	1.141
4	1.143	1.166
5	1.138	1.165
mean d	**1.1466**	
hardness	**141.1** HV	
95 %	± 2.4	

C Sample calculations

Vickers, mild steel. Mean of the six diagonals $d = \tfrac{1}{6}(0.923 + 0.922 + 0.918 + 0.913 + 0.922 + 0.924) = 0.9203$ mm, so

$$HV = \frac{1.8544 \times 100}{0.9203^2} = \frac{185.44}{0.84695} = 218.9.$$

Brinell, mild steel at 187.5 kgf. $d = 1.0350$ mm, $D = 2.5$ mm:

$$HB = \frac{2 \times 187.5}{\pi(2.5)\left(2.5 - \sqrt{2.5^2 - 1.0350^2}\right)} = \frac{375}{7.854 \times 0.22427} = 212.9.$$

95 % confidence interval. For n indents with mean diagonal $\bar{d}$ and sample standard deviation s, the interval on $\bar{d}$ is $t_{0.025,n-1}\, s/\sqrt{n}$, and it is mapped through Equation 1 or 2 to give the band quoted on the hardness. For the 60 min Al–Cu, $n = 5$, $t = 2.776$, giving ± 2.4 HV.

Slope of area against load. With the origin included and the five measured points, $m = \sum FA / \sum F^2 = 0.004554$ mm^2/kgf, so $1/m = 219.6$.